UNIT42 · FIRMWAREX

Firmware Security Assessment

My router — test analysis

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Executive summary

This assessment recorded **17 findings** across 2 audited binaries and 6 filesystem entries. Analysis status: **complete**.

Results describe the analysis performed on this image. They do not establish that the firmware is secure or that every candidate is exploitable. Severity and confidence are separate; no CVSS score is inferred.

Critical	High	Medium	Low	Info
0	1	6	10	0

Scope and coverage

Stage	Outcome	Restriction
extract	extracted	See scope limits below
static	completed	See scope limits below
emulation	completed	See scope limits below
symbolic	completed	See scope limits below
secrets crypto	completed	See scope limits below
normalize	completed	See scope limits below

Untrusted execution: blocked_pending_mentor_approval. Raw artifact access: disabled.

- Only registered controlled images execute; other images receive static and scanner analysis.
- Unknown path components are pseudonymized to prevent proprietary-string disclosure.
- Symbolic findings are bounded candidates for human review.

- Only executed stages contribute evidence. Blocked, skipped or incomplete analysis is not a clean bill of health.
- This is an automated firmware-image assessment, not a live-device or network scan.
- Recommendations are general review guidance, not verified fixes.

Binary protection summary

Binary	Architecture	NX	PIE	RELRO	Canary	FORTIFY
/bin/service-probe	x86_64	Present	Not detected	partial	Present	Not detected
/bin/taint-demo	x86_64	Present	Not detected	partial	Not detected	Not detected

Detailed findings

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1. Memory safety candidate High

Property	Result
CWE	CWE-787
Affected file	/bin/taint-demo
Architecture	x86_64
Confidence	probable
Review status	new
Address	0x500008
Function / sink	strcpy
Detected	2026-10-19T15:28:56.456388Z

Evidence

Observation	Value
condition satisfiable	true

Observation	Value
Observed taint path	0x401136 → 0x401040 → 0x500010 → 0x40115c → 0x40116b → 0x401030 → 0x500008
Reaching condition (diagnostic expression)	[{"op":"UGE","args":[{"op":"Extract","args":[247,240, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract","args":[247,240, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":126,"bits":8}]},{"op":"UGE","args": [{"op":"Extract","args":[239,232, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract" … [excerpt; full expression in JSON export]
fault address	0x404030
overflow offset	8
stack hash	bfc322b46d82910dab6757cff82dd695801b2df2172325cb b73b4e88d0926fc2

Evidence reference: evidence:f7ee5cfb9deba3f0f8edca93793aa9b62de6cdcca899317129af0ba280087bc2. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

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2. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established

Property	Result
Function / sink	Not applicable
Detected	2026-10-19T15:28:56.562475Z

Evidence

Observation	Value
rule	credential
fingerprint	1a8e2d115afac1a767b3addf1671748c34db90aebd48ddbf2a35298a8eb74d5e
byte offset	9
byte length	19

Evidence reference: evidence:03e4db6bdd0bfc6ced1f33d10c7e89d96d696c3f064bef79c86cf600826ba5b5. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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3. Embedded secret candidate Medium

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T15:28:56.562513Z

Evidence

Observation	Value
rule	credential
fingerprint	9681497a7ac81a7f0b64cb0e451b4b3405c21ddfb924176d22e390b496c8848c
byte offset	37
byte length	24

Evidence reference: evidence:6dcf8c7539d217a6cab5e8f3fa50bf04bc4c2c168b26a67d8d4f4dfa882c6413. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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4. Embedded secret candidate Medium

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T15:28:56.562537Z

Evidence

Observation	Value
rule	private_key

Observation	Value
fingerprint	b81c632439c0f8a8bb4cb040c51fdfc6d8ba6ff2d2b4f2586143aefb3be56c05
byte offset	62
byte length	78

Evidence reference: evidence:8c8cf7edde902769d3743784aa1028206c6a9929fef1c57ccd8a89869f60f55c. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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5. Weak cryptography candidate Medium

Property	Result
CWE	CWE-328
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T15:28:56.562621Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	e375c541c0b052ac825487f43b22e5bec6fe6544ee88ed5ecc0f640dacc6f2eb

Observation	Value
byte offset	237
byte length	3

Evidence reference: evidence:55abfe7324c31508b3e818c5948cdd493411df894c9d9c2d550a4431e4b831f4. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

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6. Weak cryptography candidate Medium

Property	Result
CWE	CWE-327
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T15:28:56.562640Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	d1ffa89d96cc3bcc8c37fb129c32f22e5f0f88a71172ac751465f4dd6d5d9da3
byte offset	255
byte length	3

Evidence reference: evidence:bed61611994f12c1da4e34a2d5b85917bee997e4df2c1f532fe71414303fcc0c. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

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7. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/redacted-d17f23b1fd859998
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T15:28:56.562808Z

Evidence

Observation	Value
rule	credential
fingerprint	73890b946ba5dff1ed7af807989d69ca89f36f7862feb19096a31eff9cbc7ef0
byte offset	38
byte length	25

Evidence reference: evidence:a3a11973af668bef95582df1a0aca8f80e847fa95e96590a3526fd3ea0840d97. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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8. Memory safety candidate Low

Property	Result
CWE	CWE-676
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	strcpy
Detected	2026-10-19T15:28:52.752853Z

Evidence

Observation	Value
rule	potentially_dangerous_function_symbol
invocation proven	false
overflow proven	false

Evidence reference: evidence:9f26cf77847cb0f150b93bf6a32de0fe421d1fc84e000f539293d7be970a4c71. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

9. Memory safety candidate Low

Property	Result
CWE	CWE-676
Affected file	/bin/taint-demo
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	strcpy
Detected	2026-10-19T15:28:52.753060Z

Evidence

Observation	Value
rule	potentially_dangerous_function_symbol
invocation proven	false
overflow proven	false

Evidence reference: evidence:9f26cf77847cb0f150b93bf6a32de0fe421d1fc84e000f539293d7be970a4c71. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

10. Embedded secret candidate Low

Property	Result
CWE	CWE-798

Property	Result
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T15:28:56.561237Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	1052bae02a5a479a5bd80b1b5e914d52d74b06f7fe2ddf491b4d3eb43151c359
byte offset	511650
byte length	50

Evidence reference: evidence:3201972255bf7912d445f0d0ebd3fcf09bfee48fa2a06b0c215567c53c522c0e. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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11. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64

Property	Result
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T15:28:56.561375Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	f6e17880fe137cf064b71832f7bbaf09ee5f471c201119c26578951a32dacf25
byte offset	591137
byte length	26

Evidence reference: evidence:e82729e18b072a0357b5785e6f283429c4b06d84e47afe16daec64c49a01c1db. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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12. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new

Property	Result
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T15:28:56.561408Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	2b249b3f3ea06cb8b64920c50643530091930ecd98488da4e4d1b947d0ea2b41
byte offset	591169
byte length	26

Evidence reference: evidence:4f8e0a3abc5c8279010b4fca8e48aede19616338267973590dd1335fbcc02085. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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13. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable

Property	Result
Detected	2026-10-19T15:28:56.561434Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	7700e09418f7ba0f0b29227724b64e2bd3d0150f0698d1b79d68fa1264f9c0d9
byte offset	591328
byte length	36

Evidence reference: evidence:985f3713caa8ac5727d068ba2616aa4998bb24513397998c76a2d60e6cbbd582. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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14. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T15:28:56.561455Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	d4eed6c4f3c4d97842d7fbd1903a4a19c68974be28ad74deac6292695b6c3eb4
byte offset	591392
byte length	36

Evidence reference: evidence:b49a0e82a454b4f106209a381c1cfb3f17f2634e90e0d986dab8f681c3cbb93f. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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15. Embedded secret candidate Low

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T15:28:56.562563Z

Evidence

Observation	Value
rule	certificate

Observation	Value
fingerprint	351327d669cdb009d08c2637332df0360201e57408d453cd5ed9f6c3f48a8d67
byte offset	141
byte length	75

Evidence reference: evidence:512ae0838d81ec806917f007ff11a08c7cd768dea8880eb8cb39c14420b4e1bf. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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16. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T15:28:56.562583Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	6273bdb8408b87fe580fcec9f8195c0c6d2afa398e122260de53c7b462dfa8bd

Observation	Value
byte offset	29
byte length	32

Evidence reference: evidence:eafb8f37b40d768d7b7177d68bf1583140d71936313895ca0da78f0c0a503652. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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17. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T15:28:56.562602Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	f3473c4f7f09b90e019d9827217d93be32c386e78832e79f368b47a4b751c004
byte offset	259
byte length	29

Evidence reference: evidence:d2ad1df4bad1c09ba0b17b7236083c7deb6d75580fe9e65531d7aa77a8248ec4. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

Assessment identity

Property	Value
Firmware ID	4c76a836-f47b-4c02-912c-2e64084324e3
SHA-256	a71d1e4937d2eb468ea965462c27f6b3c99eb2207fa39bd59d04161bf45856c9
Job ID	36e03525-050f-49e4-b64e-75915890e999
Image bytes	339968